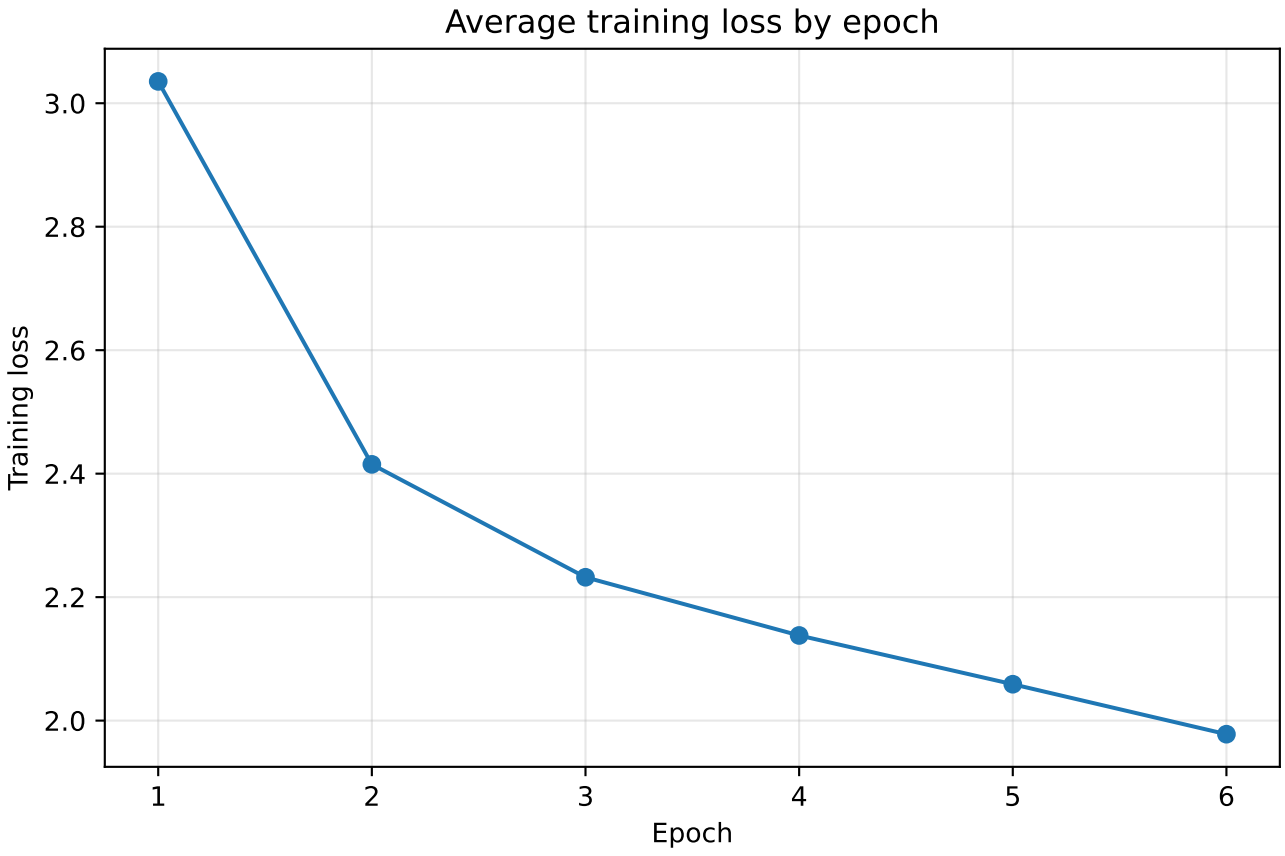
HW4: Mini GPT from Scratch

SID4: 1980 Seed: 1980
Vocabulary: 49 characters Sequence length: 128
Hidden dimension: 128 Heads: 4
Decoder blocks: 2 Parameters: 425,521
Optimizer: Adam Learning rate: 0.0003
Batch size: 32 Epochs: 6

Training loss decreased from 3.0354 to 1.9780.



Decoding comparison

Prompt: ROMEO:

Greedy:
'ROMEO:\nRomeor the that that that t t the the theee thy whe thee whe whe the the the woullll we whe woul w'

Temperature 0.5:
'ROMEO:\nHe tat me\nBy f Romeome t t neout thathe t har nor nome whe at nouthout nd the or ame ir thame n am'

Temperature 1.0:
'ROMEO:\nHe but me\nAnof RO:\nWhany t ne, owhome be thar ne\nJULJULIET:\n\n: buthy at nou t ort normertoue\nRomet'

Temperature 1.5:
'ROMEO:\nHepukesme\nAn f RO:\nRO.\nI t nenk wholf bu’ htha ny paJo ’:\nO,.: iomescat nomeg orf ntBye\nAome\nu met'

Top-k 3:
'ROMEO:\nRomeat theor thou an nore nousere theatind me wher t win t t t t at worelor amer nore ther whouthe'

Top-k 20:
'ROMEO:\nRLI now te bat sht t’onor nonotheat s: ise at inoume;\nRoff anthy woffather ell be, h’d t.\nWhamere '

Findings
Greedy decoding repeatedly chose common fragments such as 'the' and 'whe'.
Temperature 0.5 remained repetitive. Temperature 1.0 produced some
recognizable dialogue cues. Temperature 1.5 introduced the widest range
of character combinations and punctuation, but was less readable.
Top-k 3 repeated fragments; top-k 20 allowed more varied output.

Most coherent in these samples: temperature 1.0, because it includes
recognizable speaker-like text, although the passage is still fragmented.
Most diverse: temperature 1.5, at the cost of readability.

Limitation: the small training text and six training epochs limit
the model's ability to produce meaningful dialogue.